

TAZHONG, China

WMO: 517470

Lat: 38.9676N

Lon: 83.6597E

Elev: 1100

StdP: 88.79

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.1	-20.9	-28.0	0.3	-19.1	-26.0	0.4	-16.4	5.3	-1.7	4.7	-2.2	0.2	200	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	39.0	16.6	37.5	16.2	36.1	15.8	18.0	29.7	17.5	30.6	17.0	31.4	3.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	12.5	20.3	13.9	11.4	21.2	12.3	10.2	22.1	55.4	28.9	53.5	30.5	52.0	31.1	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.0	6.0	-24.7	41.3	2.5	1.3	-26.5	42.2	-27.9	42.9	-29.4	43.7	-31.2	44.6		
			-25.0	19.2	2.4	1.4	-26.8	20.1	-28.2	20.9	-29.6	21.7	-31.4	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-9.9	-2.7	7.8	15.8	21.0	25.8	27.8	26.7	20.5	11.5	1.3	-7.6
	DBStd	13.58	4.06	5.16	4.60	4.15	3.50	2.96	2.60	2.82	3.05	4.08	4.18	3.33
	HDD10.0	1916	617	357	97	6	0	0	0	0	0	30	262	547
	HDD18.3	3448	876	590	326	97	17	0	0	0	13	213	511	806
	CDD10.0	2486	0	0	29	180	340	474	551	518	315	77	1	0
	CDD18.3	976	0	0	0	21	98	224	293	259	78	2	0	0
	CDH23.3	14792	0	0	28	538	1552	3116	4214	3817	1366	161	0	0
	CDH26.7	7996	0	0	3	185	698	1747	2517	2234	578	33	0	0
Wind	WSAvg	2.2	1.1	1.6	2.2	2.8	3.1	3.1	3.2	3.0	2.5	1.8	1.3	1.1
Precipitation	PrecAvg	26	0	0	0	1	4	9	6	3	1	1	0	0
	PrecMax	46	3	1	1	10	23	24	13	17	8	4	2	1
	PrecMin	8	0	0	0	0	0	0	1	0	0	0	0	0
	PrecStd	11	1	0	0	2	7	9	4	4	2	1	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	14.6	26.8	33.8	35.9	39.2	41.5	40.1	35.2	29.7	19.9	7.3
		MCWB	-1.1	4.3	9.5	12.8	14.0	16.0	17.1	17.0	14.8	12.2	7.2	0.3
	2%	DB	2.8	12.1	23.7	30.9	34.1	37.6	39.6	38.3	33.3	27.3	16.1	4.4
		MCWB	-2.7	2.9	8.3	11.8	13.6	15.7	16.9	16.6	14.1	11.3	5.2	-1.3
	5%	DB	0.4	9.9	21.4	28.8	32.6	36.1	38.0	37.0	31.9	25.0	13.6	2.0
		MCWB	-4.1	1.7	7.4	11.0	13.1	15.3	16.6	16.1	13.8	10.2	4.3	-2.8
	10%	DB	-1.7	7.5	19.1	26.6	30.9	34.7	36.3	35.6	30.3	22.5	11.0	-0.2
		MCWB	-5.4	0.4	6.3	10.1	12.5	15.0	16.3	15.8	13.4	9.1	3.1	-4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	4.5	9.7	13.1	15.2	18.1	18.8	18.7	16.2	12.6	7.3	0.5
		MCDB	5.4	13.9	25.7	32.1	30.6	25.0	30.4	28.7	28.9	28.3	19.7	7.3
	2%	WB	-2.6	3.2	8.5	12.2	14.3	17.3	18.1	17.9	15.4	11.5	5.4	-1.3
		MCDB	2.3	11.5	22.8	30.1	31.0	26.8	30.6	30.2	28.6	26.3	15.2	4.2
	5%	WB	-3.9	1.8	7.5	11.2	13.7	16.7	17.6	17.3	14.6	10.5	4.4	-2.7
		MCDB	0.2	9.7	20.9	27.7	30.1	28.1	31.3	31.2	28.7	24.4	13.2	1.7
	10%	WB	-5.3	0.5	6.4	10.3	13.0	16.2	17.2	16.7	13.8	9.3	3.2	-3.9
		MCDB	-1.7	7.3	18.8	25.7	28.8	29.1	32.1	31.6	28.3	22.0	10.8	-0.3
Mean Daily Temperature Range	5% DB	MDBR	16.5	17.0	18.0	17.4	16.0	14.3	14.4	15.4	17.7	20.2	18.7	16.6
		MCDBR	18.6	19.5	21.7	21.3	20.3	18.3	18.7	18.8	20.8	22.1	21.7	19.0
		MCWBR	14.0	12.3	11.4	9.1	7.1	5.1	4.9	5.4	7.1	9.8	12.9	14.0
	5% WB	MCDBR	18.0	18.8	20.6	19.4	16.4	12.7	13.7	14.5	17.4	21.3	20.8	18.3
		MCWBR	13.7	11.9	10.9	8.2	5.7	3.4	3.4	4.0	5.8	9.4	12.4	13.6
Clear-Sky Solar Irradiance	taub	0.348	0.417	0.515	0.599	0.632	0.663	0.632	0.644	0.608	0.478	0.397	0.357	
	taud	2.054	1.887	1.670	1.519	1.514	1.507	1.583	1.551	1.577	1.766	1.939	2.024	
	Ebn at Noon	787	772	731	700	690	668	685	659	644	692	713	737	
	Edh at Noon	119	163	225	279	286	288	265	267	245	182	133	115	
All-Sky Solar Radiation	RadAvg	2.74	3.68	4.96	6.00	6.79	6.92	6.72	6.22	5.49	4.48	3.18	2.51	
	RadStd	0.32	0.21	0.25	0.30	0.30	0.32	0.27	0.37	0.16	0.15	0.13	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	-0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-139	-117

Nomenclature: See separate page